Name:
ID:

CSE1100 – Introduction to Computer Engineering
Quiz 1 - 11/10/2019

1. Assume you will use **1 byte (8-bits)** to represent integer numbers. Fill in the table below with the binary representations of -38 and 38 in the given format:

	Unsigned	Signed-magnitude	1's complement	2's complement
-38	none	10100110	11011001	11011010
38	00100110	00100110	00100110	00100110

2. What is the **minimum number of bits needed to represent** the given decimal numbers in **2's complement binary form**?
   a. 247 9

   b. -385 10

3. Convert the following **2's complement** binary numbers into decimal and hexadecimal.
   a. 00101011
   Decimal: 43 **Hexadecimal: x2B**

   b. 1100010
   Decimal: -30 **Hexadecimal: x62**

4. Compute the following operations. Assume each operand is a 2's complement binary number. Which operation(s) generate an **overflow**? Why?
   a. 1011 + 1101
   = 1000 (no overflow since (-5) + (-3) = -8 and we can represent it with 4 bits)

   b. 1010 – 0110
   = 0100 (overflow since (-6) – 6 = -12 and we cannot represent it with 4 bits)

   c. 01011 + 01101
   = 11000 (overflow since (11) + (13) = 24 and we cannot represent it with 5 bits)

5. Convert the following hexadecimal representations of 2's complement binary numbers **to decimal numbers**.
   a. x82C
   1000 0010 1100
   = -2048 + 32 + 8 + 4 = -2004
   b. x6A
   0110 1010 OR 110 1010
   = 64 + 32 + 8 + 2 = 106 = -64 + 32 + 8 + 2 = -22